

Katravath Srinu

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EDUCATION

Indian Institute of Technology (BHU), Varanasi

Varanasi, India

B.Tech in Computer Science and Engineering

Expected May 2027

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, AI, Machine Learning, Generative AI, Probability & Statistics, Distributed Systems
- Secured AIR 11,227 in JEE Advanced 2023 among 150,000+ candidates

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: Streamlit, LangChain, Scikit-learn, Node.js, Express.js, Pillow

AI / ML: LLMs, NLP, RAG, Prompt Engineering, Local Model Inference, AI Image Generation

Backend & APIs: REST APIs, WebSockets, Concurrent Session Management, Modular API Architecture

Cloud & DevOps: AWS (EC2, Lambda, API Gateway, S3), Docker, CI/CD Pipelines, Linux

Databases: MongoDB, MySQL, Pinecone (Vector DB)

Core CS: Concurrency & Multi-threading, Distributed Systems, Object-Oriented Design, Algorithm Design & Complexity Analysis

Tools: Git, Postman, Wireshark, Agile/Scrum

EXPERIENCE

Independent Open Source Contributor

Remote

Backend Software Engineer — Multiple Repositories

Jan 2025 – Mar 2026

- Contributed 150+ pull requests improving backend functionality, API performance, and code quality across production repositories; collaborated with distributed teams via Git workflows including feature branching, code review, and issue tracking.
- Implemented concurrent request handling and thread-safe data access patterns; optimized REST API response logic to reduce latency, improve throughput, and prevent regressions under high-load conditions.

PROJECTS

Enterprise AI Platform | *Python, Streamlit, LangChain, LLMs, AWS EC2, REST APIs*

2026

- Architected and deployed a production-grade multi-module AI platform on AWS EC2 with an LLM-powered chat assistant, AI image generation, analytics dashboard, and a prompt-to-website builder generating sites in under 10 minutes.
- Designed modular architecture with concurrent session management, multi-threaded request processing, and a decoupled REST API layer; applied complexity analysis to optimize latency and throughput across simultaneous users.

VoIP Intrusion Detection System | *Python, SIP, PCAP, Wireshark, Scikit-learn*

2025

- Built a real-time network security pipeline achieving 94% attack classification accuracy across SPIT, REGISTER flooding, and SIP spoofing; analyzed 10+ GB of PCAP traces to identify adversarial patterns.
- Engineered packet-level feature extraction reducing false positives by 40%; applied $O(n \log n)$ algorithm optimization for efficient detection on high-throughput concurrent network data streams.

Real-Time Chat Application | *Node.js, Express.js, WebSockets, MongoDB*

2024

- Built a scalable multi-user messaging platform with WebSocket-based concurrent connection handling and multi-threaded event processing for low-latency real-time communication across thousands of simultaneous users.
- Implemented RESTful authentication with persistent MongoDB storage; architected for horizontal scalability on distributed infrastructure to support enterprise-level concurrent loads.

ACHIEVEMENTS & CERTIFICATIONS

- **AWS Cloud Practitioner Essentials** — AWS Training & Certification Completion Certificate May 2026
- **HackerRank** Software Engineer Intern Certification — Data Structures, Algorithms, SQL, Python Jan 2026
- Solved **300+ DSA problems** on LeetCode & Codeforces; Codeforces Max Rating: **1350**
- Active open-source contributor with **150+ merged pull requests** across multiple production repositories